

# Mid Project Review

Link to github:

[https://github.com/TorresZ314/EE344ML\\_homework/blob/main/homework/finalProject/344Final.ipynb](https://github.com/TorresZ314/EE344ML_homework/blob/main/homework/finalProject/344Final.ipynb)

## Progress Summary

- **Data collection and preprocessing completed**
  - Utilized a dataset consisting of 63 numerical features, representing the 3D coordinates (x, y, z) of 21 distinct hand landmarks.
  - Standardization:
    - Upon plotting the frequency distributions of sampled features, we observed approximately normal (Gaussian) profiles. Therefore, we utilized StandardScaler to center and scale the data, preventing features with larger variances from disproportionately influencing the algorithms.
- **Feature engineering steps**
  - **Dimensionality Reduction:** Applied Principal Component Analysis (PCA) to compress the highly correlated 63 features down to 10 core principal components, significantly reducing computational complexity.
- **Baseline model implemented**
  - Successfully implemented and trained robust baseline classifiers, specifically Random Forest and Logistic Regression, using the full 63-dimensional feature set.
- **Initial experiments and evaluation metrics**
  - The baseline models performed exceptionally well on the original data, indicating a high signal-to-noise ratio in the landmark coordinates.
  - **Random Forest:** Achieved a Test Accuracy of **99.45%** and a Test F1 Score of **0.9938**.
  - **Logistic Regression:** Demonstrated excellent linear separability, achieving near-perfect training accuracy (**~99.76%**).
  - Conducted comparative experiments training new models on the 10-feature PCA dataset to evaluate the trade-off between dimensionality reduction and predictive performance.
- **Visualization/dashboard components**
  - Developed a custom visualization pipeline utilizing **t-SNE** to map the 63D feature space down to 2 dimensions.
  - The resulting scatter plots successfully validate the dataset's quality by displaying clearly distinct, well-separated clusters for the "Left" and "Right" gesture classes.
- **Any challenges encountered**
  - **PCA Accuracy Trade-off:** Observed a minor decrease in accuracy when switching from 63 features to 10 PCA components. Because the baseline Random Forest was generalizing perfectly without overfitting, PCA's lossy compression was subtle. We choose to continue to use a PCA reduced dataset because it reduces complexity significantly.

